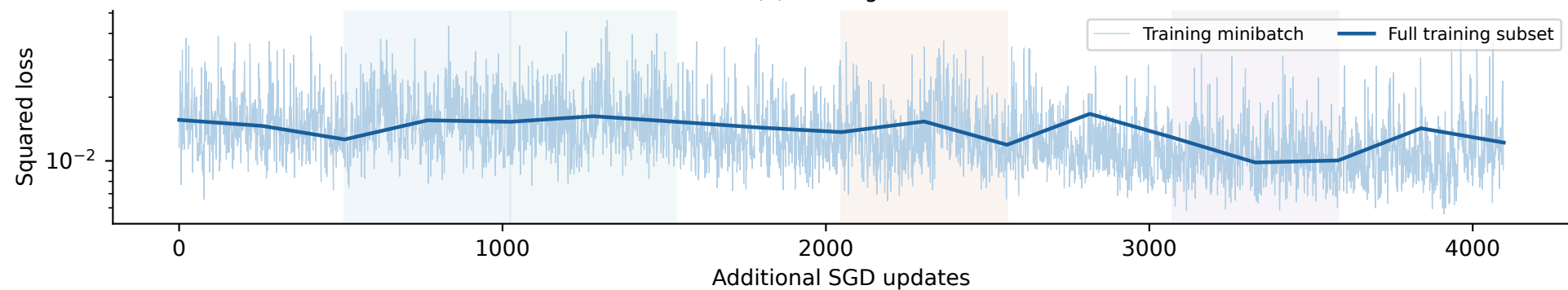
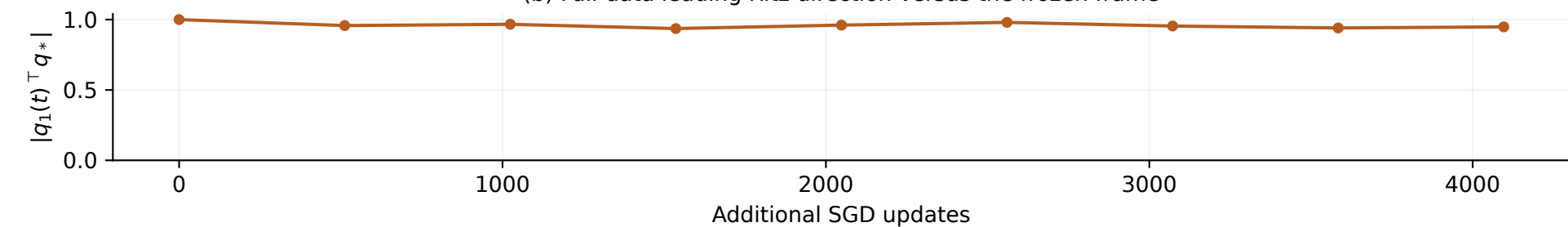


CIFAR-10 · SiLU MLP · B=32,  $\eta=0.04$   
 Checkpoint 20000 · IID minibatch SGD · status: completed

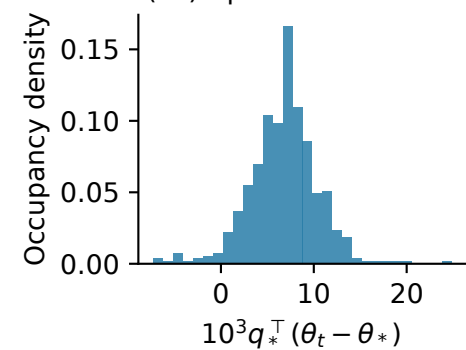
(a) Training loss



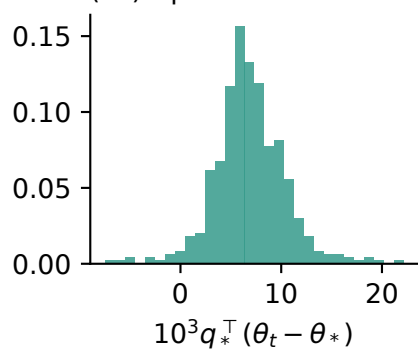
(b) Full-data leading Ritz direction versus the frozen frame



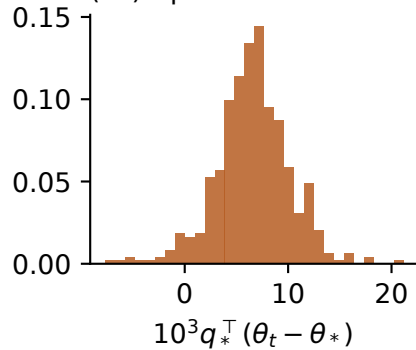
(c1) Updates 512-1023



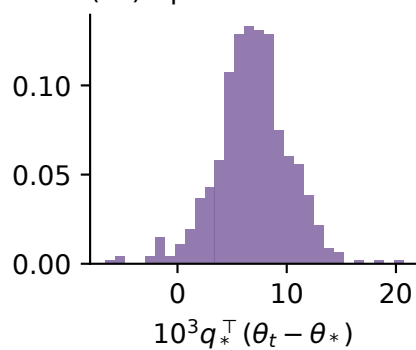
(c2) Updates 1024-1535



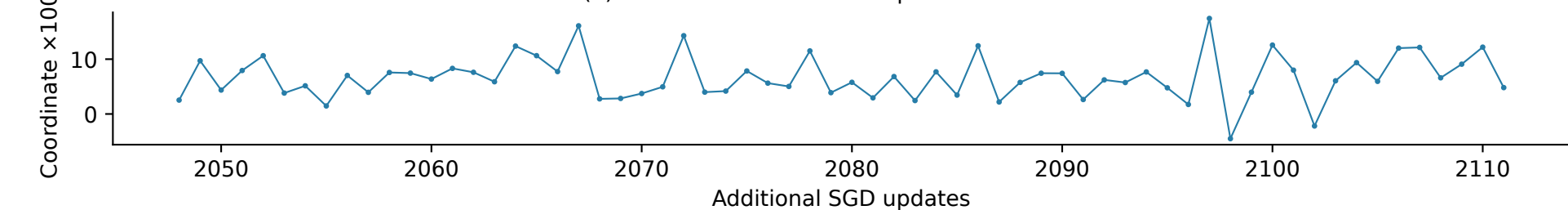
(c3) Updates 2048-2559



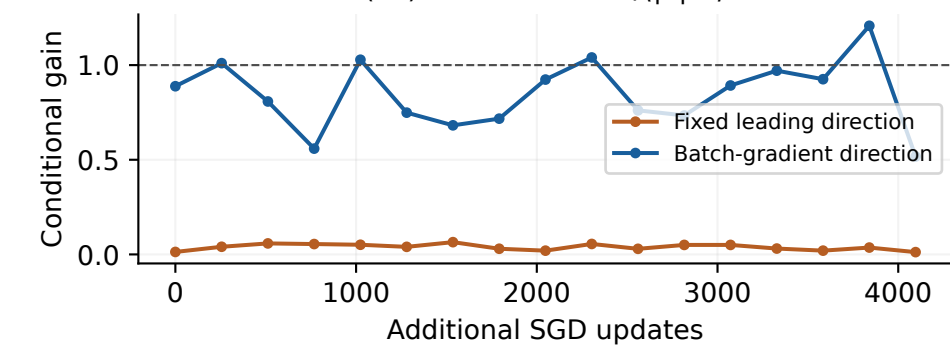
(c4) Updates 3072-3583



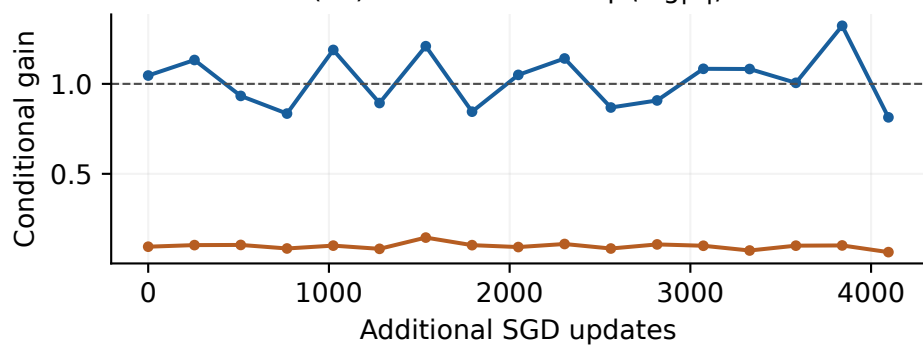
(d) Fixed-coordinate detail: updates 2048-2111



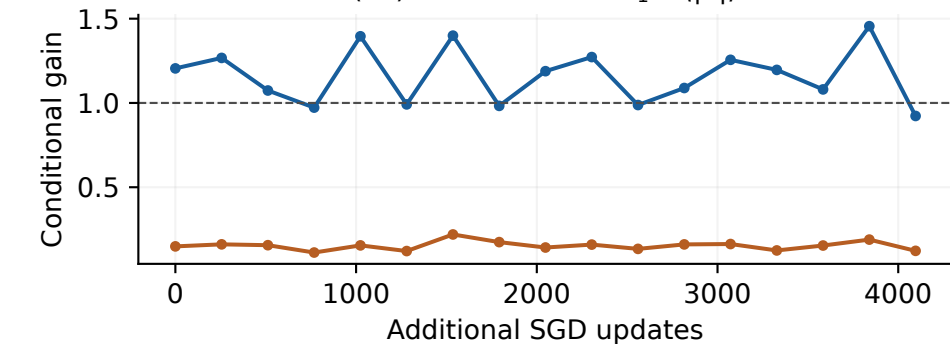
(e1) Harmonic  $H = 1/(\|a\|^{-1})$



(e2) Geometric  $G = \exp(\log\|a\|)$



(e3) Mean absolute  $A_1 = \langle \|a\| \rangle$



(e4) RMS  $A_2 = \sqrt{\langle \|a\|^2 \rangle}$

